

EXTENDED ADHERENCE AND PERSISTENCE TO HIV PREP IN A MULTIDISCIPLINARY PREP CLINIC

KRISTEN WHELCHER, PHARM.D, CSP¹, AUTUMN D. ZUCKERMAN, PHARM.D, BCPS, AAHPV, CSP¹, JOSH DECLERCQ, MS², LEENA CHOI, PHD²; SHAHRISTAN RASHID, PHARM.D CANDIDATE³, SEAN G. KELLY, MD⁴

¹VANDERBILT SPECIALTY PHARMACY, VANDERBILT UNIVERSITY MEDICAL CENTER, ²DEPARTMENT OF BIOSTATISTICS, VANDERBILT UNIVERSITY MEDICAL CENTER, ³DEPARTMENT OF PHARMACY, TRISTAR CENTENNIAL MEDICAL CENTER, ⁴DEPARTMENT OF MEDICINE, VANDERBILT UNIVERSITY MEDICAL CENTER

VANDERBILT UNIVERSITY
MEDICAL CENTER

726 McKee Avenue Nashville, TN 37211
Email: kristen.welcher@vumc.org
Tel: 615.875.8131 Fax: 615.875.0686

BACKGROUND

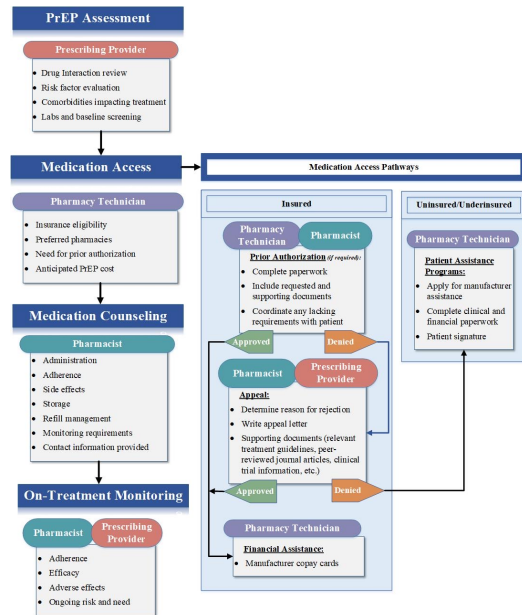
Human immunodeficiency virus (HIV) Pre-Exposure Prophylaxis (PrEP) significantly reduces the risk for HIV infection in high-risk adults

Reported HIV PrEP persistence rates are generally low at 12 to 24 months in United States PrEP clinics

Methods to identify and address barriers to HIV PrEP persistence are needed to improve low PrEP persistence rates

Objective: Describe PrEP medication adherence and persistence in patients seen at a multidisciplinary PrEP Clinic

Figure 1. Specialty Pharmacist Role in Outpatient PrEP Clinic



METHODS

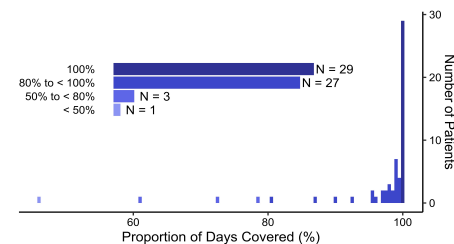
Design	Single-center, retrospective cohort
Sample	Adult patients initiating PrEP with emtricitabine-tenofovir disoproxil fumarate from a multidisciplinary clinic with prescriptions filled by Vanderbilt Specialty Pharmacy
Study Period	September 2016 - March 2019
Primary Outcome	Adherence (measured by proportion of days covered (PDC)) for the study period and persistence (measured using patient-reported discontinuation date or date of last fill plus the fill's days' supply)
Secondary Outcomes	Side effects and reasons for treatment discontinuation

Table 1. Patient Characteristics at Baseline (n=63)

Characteristic	N (%)
Age at PrEP start (years; median (IQR))	38 (29-47)
Gender, male	61 (96.8)
Race	
White	53 (84.1)
Black	5 (7.9)
Other/Unknown	5 (7.9)
Insurance type	
Commercial	59 (93.7)
Medicaid	3 (4.8)
Tricare	1 (1.6)
Indication for PrEP	
Men who have sex with men at high risk	61 (96.8)
Serodiscordant heterosexual contact	2 (3.2)
Number of sexual partners in last 6 months	
1	13 (21)
2-5	21 (33)
6-10	7 (11)
>10	8 (13)
Not reported	14 (22)
Reported condom use	
Inconsistent (<100%)	28 (60.3)
Consistent (100%)	14 (22.2)
No condom use	5 (7.9)
Not reported	5 (7.9)
Not sexually active	1 (1.6)
eGFR ≥ 60 mL/min	63 (100)
Hepatitis B status	
Susceptible at baseline	33 (52.4)
Immune due to vaccination	27 (42.9)
Immune due to natural infection	2 (3.2)
Indeterminate (isolated cAb positive)	1 (1.6)

RESULTS

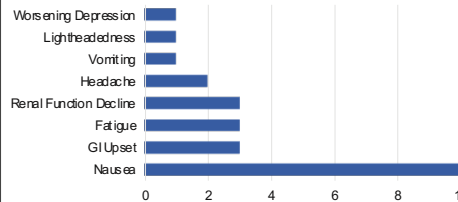
Figure 2. Adherence by PDC (n=60)



Reasons for PDC < 80%

- Held for 4 months due to IBS exacerbation not related to PrEP
- Held for 3 months pending reinstatement of insurance
- Held for 1 month due to nausea
- Patient filled PrEP at 2 different pharmacies

Figure 3. Side Effects (n=24)

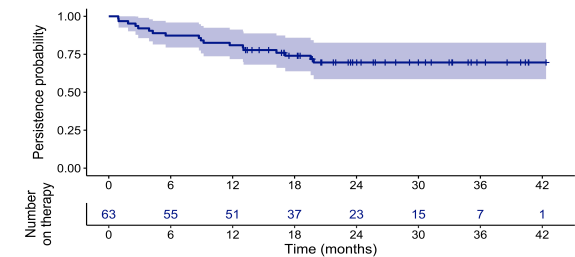


- 257 assessments were conducted with the 63 patients during the study
- 15 patients reported a total of 24 side effects

CONCLUSIONS

- Patients receiving PrEP in a multidisciplinary clinic with an integrated clinical pharmacist had high rates of adherence and persistence
- Patients reported few side effects and reasons for discontinuation were appropriate

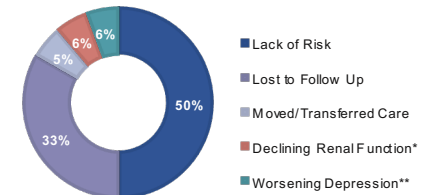
Figure 4. Persistence (n=63)



- Patients were enrolled continuously throughout the study period, therefore the length of possible follow up time is different for each patient
- Tick marks indicate patient censoring due to the end of the study period being reached

Months on Therapy	Patients on Therapy	Therapy Discontinuations	Total Discontinuations	Patients Censored	Persistence Probability	95% Confidence Interval
6	55	8	8	0	87%	80-96%
12	51	4	12	0	81%	72-91%
18	37	4	16	10	74%	64-86%

Figure 5. Reasons for Discontinuation (n=18)



*Patient with CKD resulting from DM

**Patient restarted PrEP later due to HIV exposure and continues to do well on PrEP

IQR, interquartile range; cAb, core antibody

Coy KC, Hazen RJ, Krishnam HS, Dalpiaz A, Sengler AJ. Persistence on HIV preexposure prophylaxis medication over a 2-year period among a national sample of 7148 PrEP users, United States, 2015 to 2017. *J Int AIDS Soc*. 2022;19(25):e25252. doi:10.1002/jad.25252